Chapter 4 — Exploring Numerical Variables

Exploratory Data Analysis (EDA) · Part II: Univariate Analysis

Purpose: Understand numerical variables through quantitative summaries, distribution analysis, and visualizations. This chapter focuses on investigation—not data transformation or model training.

Learning objectives

• Distinguish discrete and continuous variables from identifiers that merely look numeric.

• Calculate and interpret mean, median, mode, variance, standard deviation, quantiles, range, and IQR.

• Describe skewness and possible multimodality.

• Use histograms, boxplots, and ECDFs to explore distributions.

• Investigate unusual observations without automatically deleting them.

• Build a reusable numerical-summary function in Pandas.

1. Review of earlier concepts

Chapter 1 — What Is EDA?

EDA systematically investigates data structure, quality, patterns, and limitations before conclusions or modeling. Ask: What data do I have? Can I trust it? What patterns or relationships does it contain? Quantitative and visual EDA complement one another.

Chapter 2 — Understanding the Dataset

Establish what each row represents, what each column measures, and the dataset's granularity. Income is ambiguous without knowing whether it is monthly or annual, which currency is used, and whether it describes an individual or household.

Review of Chapter 3 — Data Quality Assessment

Check missingness, implausible values, data types, duplicate records, inconsistent units, and domain constraints. An age of 999 is numerically valid but probably invalid in a customer dataset.

• Quantify missing values; distinguish missingness from a valid zero.

• Check duplicate rows and, where appropriate, duplicate keys.

• Check valid ranges and cross-column consistency.

• Distinguish confirmed errors from suspicious observations.

• Document findings rather than silently changing the data.

Scope boundary: EDA identifies and investigates quality problems. Imputation, outlier removal, and transformations belong to preprocessing or feature engineering and are outside this chapter.

What the Chapter 3 coding project should teach you

A reusable audit should answer: Where are potential problems? How common are they? Which findings need investigation? This chapter builds on that audit.

2. Preview of this chapter

A dataset can look ordinary in its first five rows while hiding a long income tail or an unusual age. A systematic workflow is: establish meaning and units; count valid and missing observations; calculate center; measure spread; inspect quantiles and shape; visualize; investigate unusual observations; and interpret cautiously.

3. What is a numerical variable?

A numerical variable represents a quantity for which numerical operations have meaningful interpretations. Age, income, distance, temperature, weight, and number of purchases are examples. Customer IDs, postal codes, phone numbers, and account numbers generally identify entities rather than measure quantities.

Discrete and continuous variables

Discrete variables take distinct, countable values, such as number of orders or support tickets. Continuous variables represent measurements that can in principle take any value within an interval, such as height, weight, distance, duration, or temperature. Recorded measurements may be rounded to instrument precision.

Column	Example	Typical role
Age	32	Numerical measurement
Number of orders	7	Discrete numerical
Annual income	720000	Numerical measurement
Customer ID	1042	Identifier
Postal code	1207	Geographic identifier

Rule: Ask what the numbers mean before calculating with them.

4. Start with a trustworthy numerical column

The following example is illustrative. Customer ID is stored as a string because it identifies a record.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.DataFrame({
    "Customer ID": [101,102,103,104,105,106,107,108,109,110],
    "Age": [22,25,28,29,31,32,34,36,41,75],
    "Annual Income": [240000,280000,310000,330000,350000,
                      360000,390000,420000,510000,1800000],
    "Orders": [1,2,2,3,3,4,4,5,6,12]
})
df["Customer ID"] = df["Customer ID"].astype("string")

print(df.shape)
print(df.dtypes)
print(df.head())
print(df.isna().sum())
```

Count observations

```
age = df["Age"]
print("Total rows:", len(age))
print("Valid observations:", age.count())
print("Missing observations:", age.isna().sum())
print("Unique observed values:", age.nunique())
```

len(series) counts rows including missing values; count() counts non-missing observations; nunique() counts distinct non-missing values by default. Also check units and the earlier data-quality audit.

5. Measures of central tendency

Mean

For x■, x■, …, x■, the arithmetic mean is x■ = (Σx■)/n. It uses every value but is sensitive to extreme observations.

```
income = pd.Series([20000, 22000, 25000, 28000, 200000])
print("Mean:", income.mean())
print("Median:", income.median())
```

A very large income pulls the mean upward. It could be genuine, erroneous, or an unusual case worth investigating.

Median

The median is the middle value after sorting. With an even number of observations, it is the average of the two middle values. For 10, 20, 30, 40, 50 the median is 30; for 10, 20, 30, 40 it is (20 + 30)/2 = 25. It is less sensitive to extremes than the mean.

Mode

The mode is the most frequent value. There may be one mode, several modes, or no unique mode if all values occur equally often.

```
values = pd.Series([2, 3, 3, 4, 4, 4, 5])
print(values.mode())
```

Measure	Strength	Limitation
Mean	Uses all values; useful in arithmetic calculations	Sensitive to extreme values
Median	Resistant to extreme values	Does not reflect exact magnitudes of every observation
Mode	Identifies the most frequent value	May be non-unique or uninformative for continuous measurements

6. Measures of spread

Minimum, maximum, and range

Range = maximum − minimum. It is easy to interpret but depends entirely on the two extreme observations.

```
age = df["Age"]
print("Minimum:", age.min())
print("Maximum:", age.max())
print("Range:", age.max() - age.min())
```

Variance

Variance measures squared deviations from the mean. Population variance divides the sum of squared deviations by N. The usual sample variance divides by n − 1, the conventional unbiased estimator under standard sampling assumptions.

```
print("Sample variance:", age.var())       # ddof=1 by default
print("Population variance:", age.var(ddof=0))
```

Pandas Series.var() uses ddof=1 by default. If age is in years, variance is in years squared.

Standard deviation

Standard deviation is the square root of variance and is expressed in the original unit. It describes dispersion around the mean but not the entire distribution.

```
print("Sample standard deviation:", age.std())
print("Population standard deviation:", age.std(ddof=0))
```

Standard deviation is not the mean absolute distance from the mean: it is based on squared deviations.

7. Quantiles and the interquartile range

Q1 is the 25th percentile, Q2 is the median (50th percentile), and Q3 is the 75th percentile. IQR = Q3 − Q1 describes the spread of the middle 50% of observations.

```
age = df["Age"]
q1 = age.quantile(0.25)
q2 = age.quantile(0.50)
q3 = age.quantile(0.75)
```

```
iqr = q3 - q1
print("Q1:", q1, "Median:", q2, "Q3:", q3, "IQR:", iqr)
print(age.describe())
```

For a numerical Series, describe() reports count, mean, standard deviation, minimum, quartiles, and maximum. It is a starting point, not a complete EDA. Quantiles are often less affected by extremes than the mean and range.

8. A reusable quantitative summary

```
def summarize_numeric(series: pd.Series) -> pd.Series:
    # Describe non-missing values; report missingness separately.
    x = series.dropna()
    if x.empty:
        return pd.Series({
            "count": 0, "missing": series.isna().sum(),
            "unique": series.nunique()
        })
    return pd.Series({
        "count": x.count(),
        "missing": series.isna().sum(),
        "unique": x.nunique(),
        "mean": x.mean(),
        "median": x.median(),
        "mode_count": len(x.mode()),
        "std": x.std(),
        "variance": x.var(),
        "min": x.min(),
        "q1": x.quantile(0.25),
        "q3": x.quantile(0.75),
        "max": x.max(),
        "range": x.max() - x.min(),
        "iqr": x.quantile(0.75) - x.quantile(0.25),
        "skewness": x.skew(),
    })
```

```
print(summarize_numeric(df["Annual Income"]))
```

The function excludes missing values from descriptive calculations while reporting missingness separately. It does not modify the original DataFrame. For production use, validate the column's numeric type and decide how to handle infinity or other non-finite values.

9. Distribution shape

Symmetry and skewness

A symmetric distribution is approximately balanced around its center; mean and median are often close but need not be exactly equal. Right-skewness means a longer or heavier tail toward larger values; left-skewness means a tail toward smaller values. Income and house prices can be right-skewed.

```
print(df["Annual Income"].skew())
```

Pandas skewness is a sample-based measure. Positive values often indicate right-skewness and negative values left-skewness, but values near zero do not guarantee symmetry. Skewness can be sensitive to extremes and small samples.

Multimodality

Several peaks may reflect subgroups or mixed processes—for example, part-time and full-time workers in one income column. Multiple peaks motivate investigation but do not prove that distinct populations exist.

10. Visualizing numerical variables

Use plots to inspect shape. The examples use Matplotlib's object-oriented API, with an explicit figure and axes.

Histogram

```
fig, ax = plt.subplots(figsize=(8, 5))
ax.hist(df["Age"].dropna(), bins=8, edgecolor="black")
ax.set_title("Distribution of Customer Age")
ax.set_xlabel("Age (years)")
ax.set_ylabel("Number of customers")
```

```
fig.tight_layout()
plt.show()
```

Histograms can reveal concentration, spread, skewness, peaks, gaps, and potential extremes. Bin choice matters: too few bins can hide structure; too many can appear noisy. Try more than one reasonable bin count.

Density histogram

```
fig, ax = plt.subplots(figsize=(8, 5))
ax.hist(df["Annual Income"].dropna(), bins=8,
        density=True, edgecolor="black")
ax.set_title("Income Distribution (Density Scale)")
ax.set_xlabel("Annual income")
ax.set_ylabel("Density")
fig.tight_layout()
plt.show()
```

With density=True, the total histogram area is approximately 1. The vertical axis is density, not the probability of a single exact value.

Boxplot

```
fig, ax = plt.subplots(figsize=(8, 3))
sns.boxplot(x=df["Annual Income"], ax=ax)
ax.set_title("Annual Income: Boxplot")
ax.set_xlabel("Annual income")
fig.tight_layout()
plt.show()
```

The box generally spans Q1 to Q3, with a line at the median. Under the common Tukey convention, whiskers extend to the most extreme observations within Q1 – 1.5 × IQR and Q3 + 1.5 × IQR. Points beyond whiskers are potential outliers under this rule—not automatically errors or values to remove.

Histogram	Boxplot
Shows overall distribution shape	Compact summary of center and spread
Can reveal peaks and gaps	Highlights potential extreme observations
Depends on bin choices	Hides many details of shape
Useful for shape exploration	Useful for comparing spread across groups

Empirical cumulative distribution function (ECDF)

An ECDF shows the proportion of observations less than or equal to each value. It can answer: What proportion of customers have annual income at or below ■400,000?

```
fig, ax = plt.subplots(figsize=(8, 5))
sns.ecdfplot(data=df, x="Annual Income", ax=ax)
ax.set_title("ECDF of Annual Income")
ax.set_xlabel("Annual income")
ax.set_ylabel("Proportion at or below value")
fig.tight_layout()
plt.show()
```

Unlike a histogram, an ECDF does not require bin selection. It is useful for comparing percentiles and proportions below thresholds.

11. Investigating unusual observations

An unusual value is not necessarily an error. It may be a valid rare case, data-entry mistake, measurement problem, or evidence of a subgroup.

IQR fences as an investigation flag

```
x = df["Annual Income"].dropna()
q1 = x.quantile(0.25)
q3 = x.quantile(0.75)
iqr = q3 - q1
lower_fence = q1 - 1.5 * iqr
```

```
upper_fence = q3 + 1.5 * iqr
flagged = x[(x < lower_fence) | (x > upper_fence)]
print(flagged)
```

This is a screening rule, not a verdict. A legitimate distribution may contain many flagged values. Do not delete or cap values just because they cross a fence.

Questions to investigate

• Is the value possible given its meaning and units?

• Could it be a typo, misplaced decimal, or incorrect unit?

• Is it plausible for this population and time period?

• Could it belong to a different but legitimate subgroup?

• Can it be checked against source records or domain knowledge?

• How sensitive are conclusions to this observation?

Document the finding, evidence, and uncertainty. Treatment decisions belong to a separate preprocessing workflow.

12. Common mistakes

• Treating IDs or postal codes as measurements because they contain digits.

• Reporting the mean without checking skewness or unusual values.

• Assuming a boxplot point is necessarily erroneous.

• Treating one histogram bin setting as objectively correct.

• Confusing sample and population variance or standard deviation.

• Ignoring missingness when reporting counts.

• Interpreting a statistic without knowing units or population.

• Treating a descriptive pattern as a causal explanation.

13. Chapter summary

• Identify a variable's meaning and type before analysis.

• Use mean, median, and mode to describe central tendency.

• Use range, variance, standard deviation, quantiles, and IQR to describe spread.

• Examine skewness and possible multimodality cautiously.

• Use histograms, boxplots, and ECDFs for complementary views.

• Investigate unusual values rather than automatically deleting them.

• Report conclusions with context, limitations, and uncertainty.

14. Assignment

Part A — Conceptual questions

• Explain discrete versus continuous numerical variables; give two examples of each.

• Why can mean and median differ substantially for income?

• Explain variance versus standard deviation, including their units.

• What does IQR tell you that range does not?

• Why is a point beyond a boxplot whisker not automatically an error?

• How can a histogram and an ECDF answer different questions?

Part B — Quantitative practice

For [12, 14, 14, 15, 18, 20, 21, 22, 24, 60], calculate count, mean, median, mode, minimum, maximum, range, Q1, Q3, IQR, sample variance, and sample standard deviation. Compare mean and median, and describe what 60 does to the summary.

Part C — Practical EDA

• Identify true numerical measurements and exclude identifiers or codes.

• For each numerical measurement, report valid/missing/unique counts, mean, median, standard deviation, quartiles, range, IQR, and skewness.

• Create a histogram, boxplot, and ECDF for one numerical variable.

• Choose an unusual observation, if any, and list three plausible explanations to investigate.

• Write a short interpretation distinguishing observations from hypotheses.

Part D — Reflection

Choose one numerical statistic you have used without much thought. Explain what it reveals, what it hides, and what additional evidence you would want before making a decision.

15. Coding project — Numerical Variable Profiler

Build a reusable notebook section that profiles numerical variables in any tabular dataset. It should be a compact, reproducible EDA tool—not an automated decision-maker.

Requirements

• Load a dataset and document what one row represents.

• Identify candidate numerical columns while excluding identifiers and non-measurement codes.

• Create a summary table with valid/missing counts, unique count, mean, median, mode count, standard deviation, variance, min, Q1, median, Q3, max, range, IQR, and skewness.

• Generate a histogram, boxplot, and ECDF for at least two suitable variables.

• Flag values beyond 1.5 × IQR fences for investigation without automatically removing them.

• Write findings about center, spread, shape, unusual values, and limitations.

• Keep the original data unchanged and state analytical choices explicitly.

Deliverables

• A clean Jupyter notebook with explanatory Markdown and reproducible code.

• A numerical summary table.

• At least six labeled plots—three per selected variable.

• A short written interpretation for each selected variable.

• A note documenting assumptions, missingness, and flagged observations.

Success criterion: Another analyst should be able to rerun the notebook and understand both what the summaries show and what they do not establish.

Preview of Next Chapter

Chapter 5 — Exploring Categorical Variables

Next, we will explore categorical data using frequency tables, proportions, category cardinality, rare categories, bar charts, and relationships between categories. We will distinguish nominal and ordinal variables and interpret category imbalance without confusing frequency with importance.